

CHAO HOU, Ph.D.

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RESEARCH INTERESTS

Biological sequence → structure ensemble → function and fitness → evolution

Representation and generative deep learning of biomolecules

AI-assisted biomolecule design

Predicting the effects of genetic variants for precision medicine

Biomolecular interaction network and subcellular localization

EDUCATION

Ph.D. in Biomedical Informatics, Peking University

Sep 2020 – Jul 2023

Advisor: Tingting Li

Thesis: Identifying phase-separating proteins and characterizing their degradation regulation with machine learning.

Bachelor of Medicine and Economics, Peking University

Sep 2015 – Jul 2020

WORKING EXPERIENCE

Postdoctoral Research Scientist, Columbia University

Sep 2023 – Present

Advisor: Yufeng Shen

Clinical Internship, Peking University Third Hospital

Aug 2018 – Jan 2019

PUBLICATIONS

(# co-first author; * corresponding/co-corresponding author)

Selected:

- Understanding Language Model Scaling on Protein Fitness Prediction.*
Hou C*, Di L, Zafar A, Shen Y*. **Nature Computational Science** (accepted). BioRxiv, Aug 2025.
- Protein Language Models Trained on Biophysical Dynamics Informs Mutation Effects.*
Hou C, Zhao H, Shen Y*. **PNAS**, Jan 2026.
- MotifAE Reveals Functional Sequence Patterns from Protein Language Model: Unsupervised Discovery and Interpretability Analysis.*
Hou C*, Di L, Shen Y*. BioRxiv, Nov 2025. (under review)
- Investigating Phase Separation Properties of Chromatin-associated Proteins Using Gradient Elution of 1,6-hexanediol.*
Zhu P#, **Hou C**#, Liu M, Chen T, Li T*, Wang L*. **BMC Genomics**, Aug 2023.
- PhaSepDB in 2022: Annotating Phase Separation-related Proteins with Droplet States, Co-phase Separation Partners, and Other Experimental Information.*

Hou C#, Wang X#, Xie H#, Chen T, Zhu P, Xu X, You K, Li T*. **Nucleic Acids Research**, Jan 2023.

6. *Systematic Prediction of Degrons and E3 Ubiquitin Ligase Binding via Deep Learning.*

Hou C, Li Y, Wang M, Wu H, Li T*. **BMC Biology**, Jul 2022.

7. *Screening Membraneless Organelle Participants with Machine-learning Models that Integrate Multimodal Features.*

Chen Z#, **Hou C#**, Wang L#, Yu C, Chen T, Shen B, Hou Y, Li P*, Li T*. **PNAS**, Jun 2022.

8. *Serum Antigenome Profiling Reveals Diagnostic Models for Rheumatoid Arthritis.*

Han P, **Hou C**, Zheng X, Cao L, Shi X, Zhang X, Ye H, Pan H, Liu L, Li T*, Hu F*, Li Z*. **Frontiers in Immunology**, Apr 2022.

9. *MloDisDB: a Manually Curated Database of the Relations Between Membraneless Organelles and Diseases.*

Hou C#, Xie H#, Fu Y#, Ma Y#, Li T*. **Briefings in Bioinformatics**, Jul 2021.

Others:

10. *Expanding the Phenotypic Spectrum of MECOM-associated Syndrome: Rare Variants are Associated with Syndromic Pulmonary Arterial Hypertension.*

Welch et al. *Journal of Medical Genetics*, 2026.

11. *PhaSeDis: A Manually Curated Database of Phase Separation–Disease Associations and Corresponding Small Molecules.*

Chen et al. *Genomics Proteomics Bioinformatics*, 2025.

12. *Investigating Variant Impact on Phosphorylation Events Driving Carcinogenesis.*

Xu et al. *Briefings in Bioinformatics*, 2023.

13. *Distinctive Network Topology of Phase-Separated Proteins in Human Interactome.*

Yu et al. *Journal of Molecular Biology*, 2022.

14. *Quantifying the Phase Separation Property of Chromatin-associated Proteins under Physiological Conditions Using an Anti-1,6-hexanediol Index.*

Shi et al. *Genome Biology*, 2021.

PRESENTATIONS

Invited & Oral Talks:

2026.03 *Evolutionary and Biophysical Modeling of Protein Fitness Landscape with Language Model.* Saeed Tavazoie Lab, Columbia University, NY.

2026.02 *Learning Biophysical Dynamics with Language Models for Protein Function and Fitness Prediction.* Department of Systems Biology, Columbia University, NY.

2025.10 *Understanding Language Model Scaling on Protein Fitness Prediction.* Retreat, Department of Systems Biology, Columbia University, NY.

2024.09 *Learning Representation of Protein Dynamic Properties with a Language Model.* Retreat, Department of Systems Biology, Columbia University, PA.

2023.07 *The Degradation Regulation of Phase Separating Proteins.* Fudan International Summer School of Life Science, Shanghai, China.

2023.06 *The Degradation Regulation of Phase Separating Proteins*. School of Basic Medical Sciences, Peking University, Beijing, China.

2023.04 *The Degradation Regulation of Phase Separating Proteins*. School of Basic Medical Sciences, Peking University, Beijing, China.

2022.12 *Targeting Disordered Degrons on Phase Separating Proteins*. Xi'an Jiaotong University, Online.

Poster Presentations:

2025.02 *Learning Biophysical Dynamics with Protein Language Models*. BPS 2025, Los Angeles, US.

2024.11 *Predicting Missense Mutation Effects with Biophysics-informed Protein Language Model*. ASHG 2024, Denver, US.

2024.07 *Learning Representation of Protein Dynamic Properties with a Language Model*. ISMB 2024, Montreal, Canada.

HONORS & AWARDS

2023 First Prize, Trainee Symposium, Fudan International Summer School

2023 Peking University Outstanding Graduate

2022 Peking University President Scholarship

2021 Peking University Doctoral Student Innovation Scholarship

2020 Peking University Junior Scholarship

2016 National Scholarship

2014 Silver Medal, National Physics Olympiad for High School Students

TEACHING & MENTORING

Rotation Ph.D. Students: Di Liu (Columbia DBMI), Aziz Zafar (Columbia DBMI)

Undergraduate Students: Jason Xie (Columbia CS), Tianhao Li (PKU), Xinyi Li (PKU), Haotai Xie (PKU)

High School Students: Kevin Liu (85th Regeneron Science Talent Search Top 300 Scholar), Ludovic Pham

PROFESSIONAL SERVICE

Journal Reviewer: EBioMedicine, PLOS Computational Biology, Protein Science

Conference Reviewer: ISMB